

理论力学作业

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2020 年 11 月 11 日

第四章

3.1.(1) 系统的 Lagrange 量为:

$$\begin{aligned} L &= \left[\frac{1}{2} m \dot{x}_1^2 + \frac{1}{2} m \dot{x}_2^2 \right] - \left[\frac{1}{2} k x_1^2 + \frac{1}{2} k x_2^2 + \frac{1}{2} k (x_1 + x_2)^2 \right] \\ &= \frac{1}{2} \dot{\mathbf{x}}^T \mathbf{m} \dot{\mathbf{x}} - \frac{1}{2} \mathbf{x}^T \mathbf{k} \mathbf{x} \\ &= \frac{1}{2} [\dot{x}_1 \ \dot{x}_2] \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + \frac{1}{2} [x_1 \ x_2] \begin{bmatrix} 2k & k \\ k & 2k \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

解久期方程:

$$|\omega^2 \mathbf{m} - \mathbf{k}| = \det \begin{bmatrix} \omega^2 m - 2k & -k \\ -k & \omega^2 m - 2k \end{bmatrix} = (\omega^2 m - 2k)^2 - k^2 = 0$$

得两个本征频率:

$$\omega_1 = \sqrt{\frac{3k}{m}}, \quad \omega_2 = \sqrt{\frac{k}{m}}$$

此即为体系的小振动频率, 其中 ω_1 对应于两质点同向运动的情况, ω_2 对应于相向运动的情况。

3.1.(2) 对于系统的平衡位置:

$$2kx_0 + kx_0 = \frac{q^2}{4\pi\epsilon_0(l_0 + 2x_0)^2}$$

则 x_0 为方程:

$$4x_0^3 + 4l_0x_0^2 + l_0^2x_0 - \frac{q^2}{12\pi\epsilon_0k} = 0$$

的正根 (分析可知该方程仅有唯一正根)。

库仑势可以展开到二阶:

$$V = \frac{q^2}{4\pi\epsilon_0(l_0 + 2x_0 + x_1 + x_2)} = V_0 - V_0 \frac{x_1 + x_2}{l_0 + 2x_0} + V_0 \frac{(x_1 + x_2)^2}{(l_0 + 2x_0)^2}$$

其中:

$$V_0 = \frac{q^2}{4\pi\epsilon_0(l_0 + 2x_0)}$$

系统的 Lagrange 量为:

$$L = \left[\frac{1}{2} m \dot{x}_1^2 + \frac{1}{2} m \dot{x}_2^2 \right] - \left[\frac{1}{2} k (x_1 + x_0)^2 + \frac{1}{2} k (x_2 + x_0)^2 + \frac{1}{2} k (x_1 + x_2 + 2x_0)^2 + V \right]$$

$$\begin{aligned}
& -2 \times \frac{1}{2} k x_0^2 - \frac{1}{2} k (2x_0)^2 - V_0 \Big] \\
& = \left[\frac{1}{2} m \dot{x}_1^2 + \frac{1}{2} m \dot{x}_2^2 \right] - \left[\frac{1}{2} k x_1^2 + \frac{1}{2} k x_2^2 + \frac{1}{2} a (x_1 + x_2)^2 \right] \\
& = \frac{1}{2} \dot{\mathbf{x}}^T \mathbf{m} \dot{\mathbf{x}} - \frac{1}{2} \mathbf{x}^T \mathbf{k} \mathbf{x} \\
& = \frac{1}{2} \begin{bmatrix} \dot{x}_1 & \dot{x}_2 \end{bmatrix} \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2k+a & k+a \\ k+a & 2k+a \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}
\end{aligned}$$

解久期方程:

$$|\omega^2 \mathbf{m} - \mathbf{k}| = \det \begin{bmatrix} \omega^2 m - 2k - a & -k - a \\ -k - a & \omega^2 m - 2k - a \end{bmatrix} = (\omega^2 m - 2k - a)^2 - (k + a)^2 = 0$$

其中:

$$a = \frac{2V_0}{(l_0 + 2x_0)^2} = \frac{q^2}{2\pi\epsilon_0(l_0 + 2x_0)^3}$$

得两个本征频率:

$$\omega_1 = \sqrt{\frac{3k+a}{m}} = \sqrt{\frac{1}{m} \left(3k + \frac{q^2}{\pi\epsilon_0(l_0 + 2x_0)^3} \right)}, \quad \omega_2 = \sqrt{\frac{k}{m}}$$

此即为体系的小振动频率, 其中 ω_1 对应于两质点同向运动的情况, ω_2 对应于相向运动的情况。

3.2 系统的 Lagrange 量为:

$$\begin{aligned}
L & = \left[\frac{1}{2} \cdot \frac{1}{3} m l^2 \dot{\phi}_1^2 + \frac{1}{2} \cdot \frac{1}{12} m l^2 \dot{\phi}_2^2 + \frac{1}{2} m \left(l \dot{\phi}_1 + \frac{l}{2} \dot{\phi}_2 \right)^2 \right] \\
& \quad - \left[m g \frac{l}{2} \cos \phi_1 + m g \left(l \cos \phi_1 + \frac{l}{2} \cos \phi_2 \right) - m g \frac{l}{2} - m g \frac{3l}{2} \right] \\
& = \frac{1}{2} \dot{\boldsymbol{\phi}}^T \mathbf{m} \dot{\boldsymbol{\phi}} - \frac{1}{2} \boldsymbol{\phi}^T \mathbf{k} \boldsymbol{\phi} \\
& = \frac{1}{2} \begin{bmatrix} \dot{\phi}_1 & \dot{\phi}_2 \end{bmatrix} \begin{bmatrix} \frac{4}{3} m l^2 & \frac{1}{2} m l^2 \\ \frac{1}{2} m l^2 & \frac{1}{3} m l^2 \end{bmatrix} \begin{bmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} \phi_1 & \phi_2 \end{bmatrix} \begin{bmatrix} \frac{3}{2} m g l & 0 \\ 0 & \frac{1}{2} m g l \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix}
\end{aligned}$$

解久期方程:

$$\begin{aligned}
|\omega^2 \mathbf{m} - \mathbf{k}| & = \det \begin{bmatrix} \frac{4}{3} \omega^2 m l^2 - \frac{3}{2} m g l & \frac{1}{2} \omega^2 m l^2 \\ \frac{1}{2} \omega^2 m l^2 & \frac{1}{3} \omega^2 m l^2 - \frac{1}{2} m g l \end{bmatrix} \\
& = \frac{7}{36} \omega^4 m^2 l^4 - \frac{7}{6} \omega^2 m^2 g l^3 + \frac{3}{4} m^2 g^2 l^2 \\
& = 0
\end{aligned}$$

得两个本征频率:

$$\omega_1 = \sqrt{\frac{21 + 6\sqrt{7}}{7} \cdot \frac{g}{l}}, \quad \omega_2 = \sqrt{\frac{21 - 6\sqrt{7}}{7} \cdot \frac{g}{l}}$$

此即为体系的小振动频率。

3.3.(1) 对于系统的平衡位置:

$$2kx_0 = mg$$

系统的 Lagrange 量为:

$$\begin{aligned} L &= \left[\frac{1}{2}m \left(\frac{\dot{x}_1 + \dot{x}_2}{2} \right)^2 + \frac{1}{2} \cdot \left(\frac{1}{12}ml^2 \right) \cdot \left(\frac{\dot{x}_1 - \dot{x}_2}{l} \right)^2 \right] \\ &\quad - \left[\frac{1}{2}k(x_1 + x_0)^2 + \frac{1}{2}k(x_2 + x_0)^2 - 2 \times \frac{1}{2}kx_0^2 - mg \left(\frac{x_1 + x_2}{2} \right) \right] \\ &= \left[\frac{1}{2}m \left(\frac{\dot{x}_1 + \dot{x}_2}{2} \right)^2 + \frac{1}{2} \cdot \left(\frac{1}{12}ml^2 \right) \cdot \left(\frac{\dot{x}_1 - \dot{x}_2}{l} \right)^2 \right] - \left[\frac{1}{2}kx_1^2 + \frac{1}{2}kx_2^2 \right] \\ &= \frac{1}{2}\dot{\mathbf{x}}^T \mathbf{m} \dot{\mathbf{x}} - \frac{1}{2}\mathbf{x}^T \mathbf{k} \mathbf{x} \\ &= \frac{1}{2}[\dot{x}_1 \ \dot{x}_2] \begin{bmatrix} \frac{1}{3}m & \frac{1}{6}m \\ \frac{1}{6}m & \frac{1}{3}m \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + \frac{1}{2}[x_1 \ x_2] \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

解久期方程:

$$|\omega^2 \mathbf{m} - \mathbf{k}| = \det \begin{bmatrix} \frac{1}{3}\omega^2 m - k & \frac{1}{6}\omega^2 m \\ \frac{1}{6}\omega^2 m & \frac{1}{3}\omega^2 m - k \end{bmatrix} = \left(\frac{1}{3}\omega^2 m - k \right)^2 - \left(\frac{1}{6}\omega^2 m \right)^2 = 0$$

得两个本征频率:

$$\omega_1 = \sqrt{\frac{6k}{m}}, \quad \omega_2 = \sqrt{\frac{2k}{m}}$$

此即为体系的小振动频率, 其中 ω_1 对应于杆发生转动的情况 (即 x_1 与 x_2 反向且大小相同), ω_2 对应于杆不发生转动的情况 (即 x_1 与 x_2 同向且大小相同)。

3.3.(2) 根据对称性, 系统有两种简振模式:

1. 杆发生转动 (即 x_1 与 x_2 反向且大小相同):

由 $\mathbf{M} = I\boldsymbol{\beta}$ 得:

$$2 \cdot kx \cdot \frac{l}{2} = \frac{1}{12}ml^2\ddot{\theta}$$

小角近似下有:

$$x = \frac{l}{2}\theta$$

代入并化简得:

$$k\theta = \frac{1}{6}m\ddot{\theta}$$

解得本征频率为:

$$\omega_1 = \sqrt{\frac{6k}{m}}$$

2. 杆不发生转动 (即 x_1 与 x_2 同向且大小相同):

由弹簧并联公式 $k = k_1 + k_2$ 得:

$$2kx = m\ddot{x}$$

解得本征频率为:

$$\omega_2 = \sqrt{\frac{2k}{m}}$$

3.4.(1) 系统的 Lagrange 量为:

$$\begin{aligned}
 L &= \left[\frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}M\dot{x}_2^2 + \frac{1}{2}m\dot{x}_3^2 \right] - \left[\frac{1}{2}k(x_1 - x_2)^2 + \frac{1}{2}k(x_3 - x_2)^2 \right] \\
 &= \frac{1}{2}\dot{\mathbf{x}}^T \mathbf{m} \dot{\mathbf{x}} - \frac{1}{2}\mathbf{x}^T \mathbf{k} \mathbf{x} \\
 &= \frac{1}{2}[\dot{x}_1 \ \dot{x}_2 \ \dot{x}_3] \begin{bmatrix} m & 0 & 0 \\ 0 & M & 0 \\ 0 & 0 & m \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} + \frac{1}{2}[x_1 \ x_2 \ x_3] \begin{bmatrix} k & -k & 0 \\ -k & 2k & -k \\ 0 & -k & k \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}
 \end{aligned}$$

解久期方程:

$$\begin{aligned}
 |\omega^2 \mathbf{m} - \mathbf{k}| &= \det \begin{bmatrix} \omega^2 m - k & k & 0 \\ k & \omega^2 M - 2k & k \\ 0 & k & \omega^2 m - k \end{bmatrix} \\
 &= m^2 M \omega^6 - 2km(m+M)\omega^4 + k^2(2m+M)\omega^2 \\
 &= 0
 \end{aligned}$$

得三个本征频率:

$$\omega_1 = 0, \quad \omega_2 = \sqrt{\frac{k}{m}}, \quad \omega_3 = \sqrt{\frac{k}{m} + \frac{k}{2M}}$$

此即为体系的小振动频率。

3.4.(2) 设三个广义坐标为:

$$q_1 = x_2 - x_1, \quad q_2 = x_2, \quad q_3 = x_3 - x_2$$

则:

$$x_1 = q_2 - q_1, \quad x_2 = q_2, \quad x_3 = q_3 + q_2$$

系统的 Lagrange 量为:

$$\begin{aligned}
 L &= \left[\frac{1}{2}m(\dot{q}_2 - \dot{q}_1)^2 + \frac{1}{2}M\dot{q}_2^2 + \frac{1}{2}m(\dot{q}_3 + \dot{q}_2)^2 \right] - \left[\frac{1}{2}kq_1^2 + \frac{1}{2}kq_3^2 \right] \\
 &= \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{m} \dot{\mathbf{q}} - \frac{1}{2}\mathbf{q}^T \mathbf{k} \mathbf{q} \\
 &= \frac{1}{2}[\dot{q}_1 \ \dot{q}_2 \ \dot{q}_3] \begin{bmatrix} m & 0 & 0 \\ 0 & M & 0 \\ 0 & 0 & m \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} + \frac{1}{2}[q_1 \ q_2 \ q_3] \begin{bmatrix} k & -k & 0 \\ -k & 2k & -k \\ 0 & -k & k \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}
 \end{aligned}$$

解久期方程:

$$\begin{aligned}
 |\omega^2 \mathbf{m} - \mathbf{k}| &= \det \begin{bmatrix} \omega^2 m - k & -\omega^2 m & 0 \\ -\omega^2 m & \omega^2(2m+M) & \omega^2 m \\ 0 & \omega^2 m & \omega^2 m - k \end{bmatrix} \\
 &= m^2 M \omega^6 - 2km(m+M)\omega^4 + k^2(2m+M)\omega^2 \\
 &= 0
 \end{aligned}$$

得三个本征频率：

$$\omega_1 = 0, \quad \omega_2 = \sqrt{\frac{k}{m}}, \quad \omega_3 = \sqrt{\frac{k}{m} + \frac{k}{2M}}$$

此即为体系的小振动频率。

可见结果与 (1) 中相同，说明体系小振动频率的值不依赖于广义坐标的选取。

3.5.(1) 由题意，该系统的等效质量矩阵和等效劲度系数矩阵为：

$$\mathbf{m} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} \omega_0^2 & 0 & 0 & 0 \\ 0 & \omega_0^2 & 0 & 0 \\ 0 & 0 & 2\omega_0^2 & -2\omega_0^2 \\ 0 & 0 & -2\omega_0^2 & 3\omega_0^2 \end{bmatrix}$$

由线性代数中的公式：

$$\det \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{C} & \mathbf{B} \end{bmatrix} = \det \mathbf{A} \cdot \det \mathbf{B}$$

解久期方程：

$$\begin{aligned} |\omega^2 \mathbf{m} - \mathbf{k}| &= \det \begin{bmatrix} \omega^2 - \omega_0^2 & 0 & 0 & 0 \\ 0 & \omega^2 - \omega_0^2 & 0 & 0 \\ 0 & 0 & \omega^2 - 2\omega_0^2 & -\omega^2 + 2\omega_0^2 \\ 0 & 0 & -\omega^2 + 2\omega_0^2 & 2\omega^2 - 3\omega_0^2 \end{bmatrix} \\ &= (\omega^2 - \omega_0^2)^2 (\omega^4 - 3\omega^2 \omega_0^2 + 2\omega_0^4) \\ &= 0 \end{aligned}$$

得四个本征频率：

$$\omega_1 = \omega_2 = \omega_3 = \omega_0, \quad \omega_4 = \sqrt{2}\omega_0$$

此即为体系的小振动频率。